Interactive Black Mirror episode
Netflix is reportedly going to release an interactive episode in season 5 of Black Mirror.
https://dgit.in/nov18-16-1

Foldable Samsung smartphone
We might see a foldable Galaxy X phone at Samsung's developer conference this November 7-8.
https://dgit.in/nov18-16-2

Understanding real-time ray tracing

The RTX way

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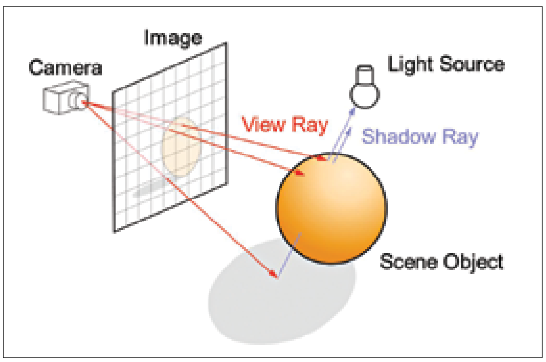
Even a cursory glance at any marketing or promotional material pertaining to the NVIDIA Turing architecture will tell you how much NVIDIA is emphasising on the real-time ray tracing aspect of the new GPUs. Ray tracing on its own, isn't a new concept. In fact, it's been around since 1968. So why is it getting popular all of a sudden? To put it in simple words, Ray-tracing was extremely resource intensive back then but the new GPUs can not only handle ray tracing easily but it can take it up a notch and trace rays in real time. Before we understand the new Ray Tracing feature present on NVIDIA Turing GPUs, let's figure out how ray tracing evolved into its current avatar.

TRACING BACK IN THE DAYS

Arthur Appel came up with the concept of what was termed as Ray Casting back then. The algorithm would trace geometric rays radiating from objects in the view cone of the observer. It was just the colour of the light that would be calculated. Moreover, there's no recursive tracing to account for rendering reflections, shadows and refractions. So rendering images with Ray Casting was possible but since the algorithm doesn't account for reflections, refractions and shadows, developers had to resort to visual trickery to make reflections and refractions happen. Texture maps were a common method to make this happen back in the day. Scanline rendering was an algorithm that was used to achieve similar results much quicker, though not as detailed as Ray Casting.

There were a few assumptions to be made in Ray Casting, one of them being that if any surface faces a source of light, it will have no shadows on it. So if we are to assume a light source, a pencil and an orange to be placed along the same line, then the pencil which sits between the light source and the orange, will not cast a shadow on to the orange. This simplification made tracing the ray of light, from a source to the viewer's eye, much simpler. There's obviously a lack of realism now that we think about it, but back then, Ray Casting was an eye opener.



Ray tracing builds an image by extending rays into a scene

Video games such as Wolfenstein 3D and the Comanche Series made use of Ray Casting algorithms. In Wolfenstein 3D, the world was built using a square-based grid of walls which were of uniform height. These merged with solid coloured floors and ceilings. While illuminating the world, a single ray is traced for every column of pixels on the screen. A vertical slice of the wall texture is then selected and scaled based on where it collides with the ray. This way the distance could be calculated and the walls could be scaled accordingly. Since the ceiling and floors are uniformly coloured, there's no need to worry about those. This also reduced the computational and memory overhead. The savings could then be utilised to render the bodies that are in motion in the open areas of the map. The Comanche Series handled Ray Casting in a slightly different manner. Individual rays are traced for each column of screen pixels and when the ray interacts with an object, it's plotted against a height map. It then determines which of the pixels are visible and which aren't, subsequently using the texture map to pick the corresponding colour for the pixel.

The next major advancement with Ray Tracing was that of Recursive Ray Tracing. Older algorithms lacked realism because they didn't account for reflections, refractions and shadows. This is because these older algorithms, as we mentioned earlier, would simply calculate the colour upon hitting an object. So the tracing process ended then and there. Turner Whitted, in 1979, decided to let the ray continue even after hitting an object. Except now, the ray had three options. It could generate up to three new types of rays, one for reflection, one for refraction and one for shadows. Reflection rays would